Brady M. Chisholm

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Research Interests

I am interested in how listeners extract meaningful signals from competing sound, and how that ability changes with hearing loss, age, or auditory prostheses. I have worked on this question in an animal model (spatial and comodulation masking release in Cope's gray treefrog), in human listeners (speech-in-speech perception and listening effort, including listeners with mild cognitive impairment), and in implant psychophysics (pitch scaling in cochlear-implant listeners and test development for an auditory nerve implant). I pair behavioral psychophysics with EEG and pupillometry, and analyze the resulting data with mixed-effects and generalized additive models in R and MATLAB. In graduate training, I want to connect these behavioral measures to neural encoding, using auditory nerve models, temporal response functions, and EEG to study how speech is represented when the auditory periphery is degraded or replaced.

Education

Bachelor of Science, Psychology *Sept. 2023–Aug. 2025*

University of Minnesota, Minneapolis, Minnesota

- Awarded scholarship and completed Dean's First-Year Research and Creative Scholars, training in Professor Andrew Oxenham's lab in data collection for EEG and auditory-based experiments, and audiogram methods
- Quantitative & computational: Regression & Correlated Data; R & Reproducible Research; Introduction to Programming Concepts (Python); Calculus I; Introduction to Physics
- Domain: Psychology of Hearing; Psychology of Attention; Research Methods

Gustavus Adolphus College, St. Peter, Minnesota *Sept. 2022–May 2023*

Coursework toward B.A.; transferred to UMN

- Varsity Men's Swim & Dive Competitor, MIAC Conference

Research Experience

Research Professional *August 2026–Present*

Sensory Optimization & Neural Implant Coding Lab, UMN Dept. of Biomedical Engineering (PI: Dr. Hubert Lim)

- Validate a six-module psychophysical test suite (MATLAB) for an in-human auditory nerve implant (ANI), verifying stimulus output and response hardware through the MED-EL research stimulation interface and Picoscope measurement prior to clinical release.
- Deliver verified protocols to the collaborating clinical team in Hannover, Germany, for administration with the first implanted ANI recipient.
- Lead recruitment and data collection for a pitch-scaling study in normal-hearing and cochlear-implant listeners, establishing comparison data for the same task in ANI recipients.
- Analyze pitch-scaling data (rank correlation, placement error in octaves) and draft the manuscript; results to be presented at the CATSS Fall Research Symposium (Nov. 2026).

Research Professional (25% appointment) *August 2026–Present*

Center for Applied Translational and Sensory Science, UMN College of Liberal Arts (Director: Dr. Andrew Oxenham)

- Administer free hearing and vision screenings aboard the See, Hear, MN mobile sensory unit, a sound-attenuating screening suite serving underserved metro and rural Minnesota communities; over 1,000 screenings completed to date.
- Operate and deploy the unit to county fairs, health festivals, food pantries, and the Minnesota State Fair, including site setup enabling data collection by partner CATSS labs.
- Counsel community members on sensory health and referral pathways for follow-up care.

Psychology Consultant *May 2026–July 2026*

Dr. Piper Ranallo, UMN Institute for Health Informatics

- Conducted a structured review of over 300 behavioral health assessment instruments, tracing each to its original development publication and subsequent validation studies.
- Built and delivered a structured database cataloging instrument provenance, psychometric validation, and source citations.

Research Professional *November 2025–July 2026*
Research Assistant *May 2024–August 2025*
Auditory Perception & Cognition Lab, UMN Psychology Department (PI: Dr. Andrew Oxenham)

- Assisted with NIH-funded research on speech-in-speech perception using EEG and behavioral experiments.
- Led a first-author analysis of pupillometric fatigue in a listening-effort paradigm using GAMMs; presented at ASA (May 2025) and CATSS (April 2026), manuscript in preparation.
- Recruited listeners with mild cognitive impairment (MCI) through M Health Fairview for EEG experiments comparing listeners with and without MCI.
- Processed, analyzed, and visualized multimodal datasets in R and MATLAB; applied GAMMs and mixed-effects modeling.

Research Assistant *May 2025–October 2025*
Noel Lab, UMN Neuroscience Department (PI: Dr. Jean-Paul Noel)

- Designed, recruited and piloted human participants (N = 11) independently in an experiment investigating visual encoding and reward prediction, managed data collection and preprocessing.
- Built analysis pipelines in R and MATLAB using signal filtering; synchronizing data.
- Organized and implemented lab-wide human participant database for HIPAA and IRB compliance.

Research Assistant *May 2024–August 2024*
Animal Communication Lab, UMN Ecology, Evolution, and Behavior (PI: Dr. Mark Bee)

- Supported NSF funded research on spatial release from masking (SRM) and comodulation masking release (CMR) in *Hyla chrysoscelis*, an animal model of signal detection in noisy, multi-talker environments (acknowledged contributor)
- Conducted two-alternative phonotaxis tests across six noise treatments factorially varying noise location (colocated vs. spatially separated) and envelope modulation (unmodulated, comodulated, uncorrelated)
- Scored response, response latency, and stimulus selection; assisted with acoustic calibration of the testing arena
- Performed animal handling, surgical preparation, and post-surgical care under IACUC protocols; collected field specimens during the spring breeding season

Research Assistant *January 2023–June 2023*
Hecht Attention Lab, Gustavus Adolphus College Dept of Psychology (PI: Dr. Lauren Hecht)

- Designed a behavioral experiment on entomophobic responses in the rubber hand illusion, using the Madagascar hissing cockroach, in collaboration with the Gustavus Biology Department.
- Wrote the literature review and IRB protocol; awarded a Swanson-Holcomb Undergraduate Research (SHURE) grant to fund the pilot.
- Recruited and ran pilot participants (N = 9); presented results as a talk at Spring Research Day.

Publications, Posters, and Presentations

Publications

1. **Chisholm, B. M.**, Johns, M. A., & Mesik, J. *Fatigue Does Not Interact with Experimental Manipulation in a Speech-in-Speech Listening Effort Task.* (Manuscript in preparation).

2. Feng, L., **Chisholm, B. M.**, Sondh, I., & Oxenham, A. J. *Pitch Scaling for Cochlear and Auditory Nerve Implant Fitting* (working title). (Manuscript in preparation).

Posters and Presentations

1. **Chisholm, B. M.**, Feng, L., & Oxenham, A. J. (2026, November). *An Efficient Pitch Scaling Method for Auditory Nerve Implant Fitting.* [Oral Presentation, upcoming]. Center for Applied & Translational Sensory Science Fall Research Symposium, University of Minnesota, Minneapolis, MN.

2. **Chisholm, B. M.**, Johns, M. A., & Mesik, J. (2026, April). *Assessing the Effect of Pupillometric Fatigue on Task-Evoked Responses in a Listening Effort Paradigm.* [Poster Presentation]. Center for Applied & Translational Sensory Science Spring Research Day, University of Minnesota, Minneapolis, MN.

3. **Chisholm, B. M.**, & Mesik, J. (2025, May). *Assessing the Effect of Pupillometric Fatigue on Task-Evoked Responses in a Listening Effort Paradigm.* [Poster Presentation]. 188th Meeting of the Acoustical Society of America, New Orleans, LA.

4. **Chisholm, B. M.**, & Hecht, L. (2023, May). *Impact of a Cockroach in the Rubber Hand Illusion.* [Oral Presentation]. Spring Research Day, Gustavus Adolphus College, St. Peter, MN.

Honors, Awards, and Scholarships

College of Liberal Arts Dean's List, *University of Minnesota* — *2024-2025*

1st Place, R Data Science Competition, *Psi Chi International Honor Society* — *Oct. 2024*

Undergraduate Research Opportunities Program, *University of Minnesota* — *Summer 2024*

Dean's First-Year Research & Creative Scholars, *University of Minnesota* — *Spring 2024*

Director's Residential Service Award, *Supportive Living Solutions* — *February 2024*

President's Faculty-Student Collaboration Grant, *Gustavus Adolphus College* — *Summer 2023*

Dean's List, *Gustavus Adolphus College* — *Spring 2023*

Swanson-Holcomb Undergraduate Research Award, *Gustavus Adolphus College* — *Spring 2023*

Eagle Scout, *Northern Star Council, Boy Scouts of America* — *Nov. 2021*

Skills & Certifications

Programming & Tools

- **Proficient:** MATLAB, R/RStudio, Git/GitHub
- **Modeling:** GAMMs, linear mixed-effects, regression, non-parametric statistics
- **R packages:** *ggplot2*, *mgcv*, *itsadug*, *lme4*, *tidyverse*
- **Familiar:** Python, HTML, LaTeX, Linux OS

Lab & Technical Skills

- EEG Data Collection and Post-Processing (BioSemi; EEGLab)
- Pupillometry (Tobii Pro Spectrum)
- Oscilloscope/Picoscope Testing
- Audiometric Threshold Testing
- Surgical Support & Animal Handling
- Laboratory Safety & Data Protocols; GCP

Certifications

- Fairview Non-employed Research Staff (human research subjects through M Health Fairview)
- Collaborative Institutional Training Initiative (CITI) Training (data practices, research privacy & ethics)
- HIPAA, FERPA, Information Security Training
- MoCA (Montreal Cognitive Assessment) Test Administration Certification for Mild Cognitive Impairment

Service & Leadership

Reviewing

Ad hoc reviewer, *Minnesota Undergraduate Research & Academic Journal* (2024-2025)

Other Professional Service

Appointed Member, AI Research Committee — *Sept 2026–Present*
University of Minnesota, Department of Psychology

- Departmental task force dedicated to ensuring the department effectively and appropriately leverages AI within the research lifecycle

Committee Lead, AI in Research Task Force *Feb 2026–May 2026*
University of Minnesota, Department of Psychology

- Led a departmental task force charged with establishing ethical guidelines for Generative AI use in the Psychology Department's research workflow.
- Coordinated group report writing on data privacy, mentorship, and AI-assisted research.

Academic Integrity Matters (AIM) Program Volunteer *Jan 2026–Present*
Office of Community Standards, UMN Twin Cities

- Assist with facilitating discussions with students who violated the University Student Code of Conduct; program aimed towards restorative justice.
- Represent University Community Members, tasked with promoting academic integrity by sharing perspectives on the impact of scholastic dishonesty beyond the classroom.

Science Outreach & Communication

Community Presentation, Orchestra Hall: Music and the Mind *Feb. 15, 2026*
UMN Auditory Perception & Cognition Lab, Minneapolis, MN

- Provided information on auditory science and research for Cochlear Implant (CI) users and broader community during a chamber concert event.

Guest Speaker, UMN CLA 1052 *Fall 2025*
University of Minnesota, Minneapolis, MN

- Served as panelist on undergraduate research pathways and opportunities for engagement; within and beyond the psychology department.

Community Presentation, Spotlight Science: The Science of Sound *Jan. 18, 2025*
Bell Museum, St. Paul, MN

- Assisted with interactive demonstrations explaining the auditory system to the public.

Additional Experience

Mental Health Advocate II *May 2023–August 2024*
Supportive Living Solutions Whittier Campus, Minneapolis, Minnesota

- Coordinated care for 40+ residents with diverse behavioral and mental health needs with a multidisciplinary clinical team.
- Audited and managed medication distribution to ensure full compliance with facility, state and federal regulations.
- Implemented improvements to documentation accuracy and continuity of care via tracking system.

Lead Copy Editor *September 2022–June 2023*
The Gustavian Weekly, St. Peter, Minnesota

- Managed a team of three copy editors in reviewing, editing, and publishing a weekly print newspaper.